

Toward mechanistic virtual immune cells

Christian M. Beusch, Brice Edelman, Chase A. Hubbard, Jovan S. Brockett, Jeffrey Skolnick & David E. Gordon



This Comment outlines molecular and technical considerations for developing mechanistically grounded virtual immune cell models.

There is growing interest in developing digital twins of cells that mimic the molecular machinery and behaviors of living systems, often referred to as ‘virtual cells’. These *in silico* models promise a predictive, systems-level view of biology that could accelerate the dissection of genotype-to-phenotype relationships, illuminate causal mechanisms of disease, enable high-throughput exploration of genetic, chemical and environmental perturbations, and ultimately inform patient-specific therapeutic strategies¹. Although the idea of a virtual cell is not new, advances in artificial intelligence and the rapid expansion of large-scale biological datasets have revived and intensified these efforts. Recent initiatives, such as the Arc Institute’s annual competition to benchmark progress toward a virtual cell², echo earlier community challenges such as the Critical Assessment of Protein Structure Prediction (CASP), which helped to drive major advances in structure prediction methods.

Prevailing virtual cell efforts often utilize gene expression readouts to represent cellular states and infer cellular behavior. When paired with perturbations, such approaches can reveal gene regulatory programs, features of signaling pathways, and phenotypic similarities among perturbations. However, gene expression measurements provide a limited view of the molecular processes that mediate cellular function. Furthermore, many cellular functions are regulated by biochemical mechanisms that operate independently of, or precede, transcriptional remodeling, underscoring the incomplete picture of cellular regulation provided by gene expression-based frameworks. Accordingly, integrating other sources of biological information, particularly biochemical data, is crucial for developing comprehensive, mechanistically grounded virtual cell models.

Among the diversity of cells in the body, immune cells are distinguished by their capacity to sense, interpret and respond to dynamic and potentially hostile environments. Whether resident in tissues or migrating through heterogeneous microenvironments, they integrate a wide array of inputs – including antigens, cytokines, danger signals, metabolic cues, cell–cell interactions and extracellular matrix contacts – to govern a multitude of functions including migration, effector function, tolerance and survival. These integrative responses are mediated not just by gene regulation, but also by biochemical, metabolic, spatial and biophysical processes. Capturing these facets of regulation in virtual cell models could help to reveal context-dependent mechanisms and prioritize candidate interventions for experimental and translational evaluation.

As immune cells integrate environmental inputs through biochemical pathways to determine cellular responses (Fig. 1a), simulating a virtual immune cell will require models that represent extracellular stimuli and dynamic biochemical transitions alongside

gene regulatory mechanisms (Fig. 1b). Consider T cell activation as an illustrative example of biochemical signal integration (Fig. 1c). Engagement of peptide–major histocompatibility complex (MHC) by the T cell receptor (TCR) triggers rapid signaling cascades driven by changes in protein structure, molecular interactions, post-translational modifications (PTMs), enzymatic activity, secondary messenger abundance, and spatial reorganization³. Co-stimulatory receptors such as CD28 reinforce activation; inhibitory receptors such as PD-1 attenuate signaling through mechanisms including phosphatase recruitment⁴; and pathways such as purinergic signaling integrate metabolic and paracrine cues from the tissue microenvironment⁵. These inputs, and the integrative biochemical circuitry they influence, shape immune responses upstream of transcription regulation.

While T cell signaling serves as an illustrative case, other examples of rapid and intermediate-timescale biochemical processes can be found across diverse immune contexts. In B cells, receptor engagement, proximal interactions, post-translational modifications, enzymatic activity and secondary messenger signaling also mediate antigen sensing and activation upstream of transcriptional regulation. In dendritic cells, antigen processing and presentation depend on biochemical machinery for proteolytic processing, vesicular routing, MHC loading and surface presentation. In neutrophils, chemoattractant, adhesion and pattern recognition receptor signaling converge to coordinate migration and antimicrobial effector functions, including phagocytosis, granule–phagosome fusion and NADPH oxidase activation.

A mechanistic virtual immune cell should represent a cell’s molecular constituents – their structures, localizations, interactions and dynamics – and how these properties change in response to environmental stimuli. Realizing such models will require advances across multiple layers of measurement and modeling that together capture how immune cells sense, process and execute responses within dynamically regulated feedback architectures. At a minimum, this challenge spans four mechanistic domains: extracellular sensing and interactions; intracellular biochemical networks and signal transduction; gene regulation; and the translation of molecular networks and dynamics into cellular functions.

First, models must capture extracellular communication and interactions. This requires comprehensive catalogs of extracellular inputs, including small molecules, lipids, proteins, macromolecular complexes, and extracellular matrix components, along with cognate receptors, receptor complexes, and broader interaction contexts. Considerable progress has been made on this front⁶, and experimental work continues to expand ligand–receptor and extracellular interaction databases while also capturing quantitative measurements of binding and downstream signaling responses. Advances in high-dimensional spatial analysis technologies are also providing insight into the tissue microenvironment and proximal biochemical inputs. Additionally, for adaptive immune cells, predictive models are being developed to determine which TCRs recognize particular peptide–MHC complexes⁷ and which B cell receptors bind specific native antigens⁸. A further

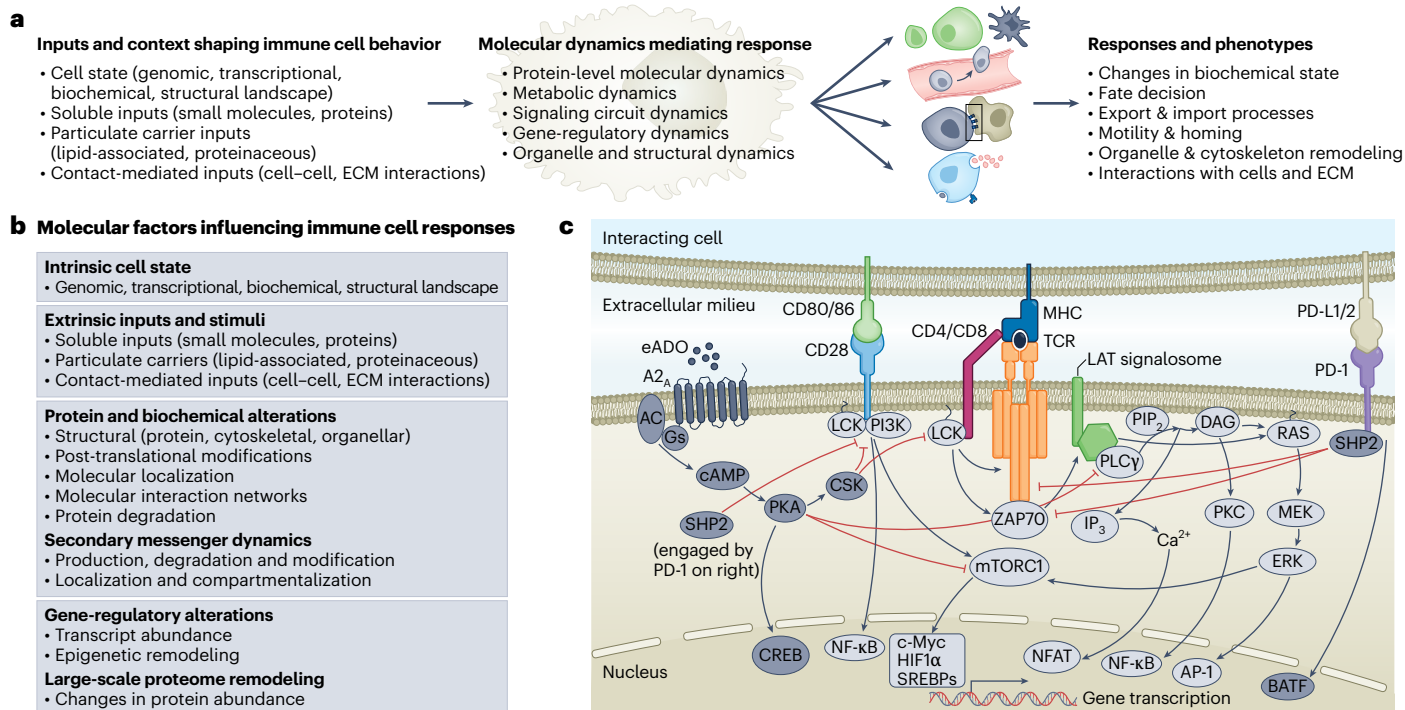


Fig. 1 | Cellular state and extracellular inputs jointly govern biochemical dynamics and integrative decision-making. **a**, Immune cells constantly integrate a wide array of inputs that shape their responses. This information is interpreted through biochemical pathways as well as epigenetic and transcriptional programs. To model a virtual immune cell, it is essential to incorporate extracellular inputs, the dynamic biochemical transitions they trigger, and their downstream consequences for cellular behavior and fate. **b**, Inputs to the cell include soluble factors, macromolecular assemblies and cell contact-mediated signals. These inputs can trigger biochemical and secondary messenger dynamics, as well as longer-term gene-regulatory and proteome remodeling programs, which collectively influence cellular behavior and fate. **c**, A highly simplified model of pathways regulating T cell activation as an example of interconnected biochemical immune signaling networks. Activating pathways include the TCR and CD28 receptor, with activating intracellular pathway components shown in light gray. Inhibitory pathways mediated by protein–protein and metabolite–protein interactions, exemplified by PD-1 and the A_{2A} receptor, are highlighted in dark gray, with inhibitory signals shown in red. The complete biochemical signaling landscape of T cells is orders of

magnitude more complex than depicted here, underscoring the need for computational approaches, including artificial intelligence, to model the integrated pathways that govern immune decision-making^{3–5,15}. A_{2A}, adenosine A_{2A} receptor; AC, adenylyl cyclase; AP-1, activator protein-1 transcription factor; BATF, basic leucine zipper transcription factor, ATF-like; cAMP, cyclic adenosine monophosphate; c-Myc, cellular myelocytomatosis oncogene; CREB, cAMP response element-binding protein; DAG, diacylglycerol; eADO, extracellular adenosine; ECM, extracellular matrix; ERK, extracellular signal-regulated kinase; G_s, stimulatory G protein; HIF1α, hypoxia-inducible factor 1α; IP₃, inositol 1,4,5-trisphosphate; LAT, linker for activation of T cells; Lck, lymphocyte-specific protein tyrosine kinase; MEK, MAPK/ERK kinase; mTORC1, mechanistic target of rapamycin complex 1; NF-κB, nuclear factor κ-light-chain enhancer of activated B cells; NFAT, nuclear factor of activated T cells; PD-1, programmed cell death protein 1; PD-L1/2, programmed death ligand 1/2; PI3K, phosphatidylinositol 3-kinase; PIP₂, phosphatidylinositol 4,5-bisphosphate; PKC, protein kinase C; PLCγ, phospholipase C γ; SHP2, Src homology 2 domain-containing protein tyrosine phosphatase 2; SREBP, sterol regulatory element-binding protein; ZAP70, ζ-chain of T cell receptor-associated protein kinase 70.

challenge is to model the biophysical properties of these interactions and understand how they shape downstream signaling.

Second, models should represent intracellular molecular networks and how they propagate and interpret biochemical signals. As immune cell signaling is dynamic and highly interconnected (Fig. 1c), models should capture how signals converge, branch and mechanistically govern cellular behavior in a temporally resolved manner. Signals are propagated through numerous mechanisms, including changes in molecular interactions, modifications and spatial distribution. Frameworks have been developed to model cellular signaling⁹. However, despite mechanistic knowledge of many immune signaling pathways, new regulatory mechanisms continue to emerge, and gaps remain in our understanding of dynamic protein interactions, PTM landscapes, molecular localization, and the diversity of cellular metabolites and

their influence on signaling networks. Ongoing experimental efforts and advances in molecular profiling, biochemical interaction mapping, imaging modalities and perturbational approaches will continue to expand the experimental evidence informing these models. Notably, perturbational studies coupled with biochemical readouts can help establish causal relationships within signaling pathways¹⁰.

Third, gene regulation should be modeled mechanistically. Predictive frameworks should incorporate chromatin state, transcription factor binding and other mechanisms governing transcriptional output, along with post-transcriptional and post-translational mechanisms that regulate RNA and protein abundance, as well as functional availability. Collectively, these regulatory layers shape not only immediate transcriptional responses but also the longer-term fate decisions that underlie immune cell differentiation and function. There is a large and

rapidly growing body of data in this realm, including multiple transcription factor binding site databases; a growing number of studies leveraging assay for transposase-accessible chromatin sequencing (ATAC-seq), chromatin immunoprecipitation sequencing (ChIP-seq) and Perturb-seq; and expanding data on RNA regulation and protein stability. Computational approaches for predicting gene regulatory interactions, including transcription factor binding site prediction¹¹ and gene regulatory network inference, also continue to improve.

Across these three domains, emerging computational tools are improving our ability to model molecular determinants of immune behavior, including protein–protein interactions, small molecule–protein interactions, nucleic acid–protein interactions, and the structural effects of PTMs and mutations^{12–14}. Yet substantial barriers remain to representing immune cells as time-resolved dynamical systems. Compounding the challenge of modeling these systems, immune cell behavior is shaped by layered positive and negative feedback, adaptation and tonic signaling that tune, constrain and stabilize cellular responses. Realistic virtual immune cells will therefore require not only reconstruction of feed-forward pathways, but also quantitative models of feedback regulation, state dependence and homeostatic control.

The fourth major challenge is understanding how molecular organization gives rise to cellular function. Immune cell behaviors emerge from complex molecular assemblies and pathways whose effects on cellular phenotypes can be difficult to predict owing to structural dynamics, the scale of molecular networks within immune cells, context-dependent regulation, and nonlinear and stochastic processes. Function is often inferred using frameworks that learn from prior biological observations and can incorporate diverse sources of information, including molecular and pathway annotations, large-scale observational and perturbation datasets, structural information and network relationships. Although these frameworks can model known conditions, their ability to generalize to unseen contexts remains limited. Immune cells integrate diverse intrinsic and extrinsic signals to produce context-dependent responses (Fig. 1a,b), and predicting these responses across the vast range of physiological conditions, genetic backgrounds and potential therapeutic strategies remains a key objective.

Therefore, at present, a fully mechanistic virtual immune cell remains an aspirational goal. Achieving greater biological realism will require addressing several major considerations, including sparse and heterogeneous measurements across scales, incomplete knowledge of underlying mechanisms, the diversity of relevant biological contexts, and the computational demands of modeling multiscale dynamics. In the near term, progress is likely to come from iterative, purpose-built models that capture specific aspects of immune cell molecular networks and behavior, rather than full digital replicas of immune cells. Even so, continued advances in multimodal molecular profiling, causal perturbation analysis, structural biology and artificial intelligence-driven approaches are steadily expanding the empirical and technical foundation needed to build, test and refine increasingly granular virtual immune cell models.

As increasingly predictive virtual immune cells take shape, their applications could span basic discovery, translational research and medicine. In fundamental immunology, such models would support

the generation of testable causal hypotheses, enable *in silico* exploration of immune behavior across diverse genetic and environmental contexts, and augment interpretation of complex multidimensional datasets. In translational research, virtual immune cells could accelerate target identification, prioritize immunomodulatory strategies and support the discovery and optimization of patient-specific therapies. In clinical and public-health settings, increasingly accurate models may ultimately improve prediction of therapeutic responses, help to stratify patients and guide the design of more precise and effective interventions. Together, these capabilities will move immunology toward a more mechanistic and actionable understanding of immune function in health and disease.

Christian M. Beusch ^{1,2,3,4,5}, **Brice Edelman**⁶, **Chase A. Hubbard**⁶, **Jovan S. Brockett**^{1,2,3,4}, **Jeffrey Skolnick** ⁶ & **David E. Gordon** ^{1,2,3,4} ✉

¹Pathology Advanced Translational Research Unit, Department of Pathology and Laboratory Medicine, Emory University School of Medicine, Atlanta, GA, USA. ²Emory Vaccine Center, Emory University School of Medicine, Atlanta, GA, USA. ³Winship Cancer Institute, Emory University, Atlanta, GA, USA. ⁴Wallace H. Coulter Department of Biomedical Engineering, Emory University and Georgia Institute of Technology, Atlanta, GA, USA. ⁵Department of Surgical Sciences, Uppsala University, Uppsala, Sweden. ⁶Center for the Study of Systems Biology, Georgia Institute of Technology, Atlanta, GA, USA.

✉ e-mail: david.ezra.gordon@emory.edu

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Competing interests

D.E.G., J.S. and J.S.B. are co-founders of VirtuCell. J.S. is a co-founder of Ziab. D.E.G. is a founding advisor for Ziab.